

Question	Marking Guidance	Mark	Comments
03.1	Membrane-bound organelle(s) OR Mitochondrion/mitochondria OR Vesicle(s)/lysosomes OR (Rough) endoplasmic reticulum OR Nucleus/(double) nuclear membrane/pore(s)/ nuclear envelope;	1 (AO1)	Ignore rER OR rough ER Accept '80S ribosomes' OR 'large(r) ribosomes' Reject smooth Reject nucleolus Reject cell membrane

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03.2	1. Modify/package/transport proteins OR Make/transport glycoproteins; 2. Modify/package/transport lipids OR Make/transport glycolipids; 3. Forms/releases vesicles/lysosomes;	2 max (2 x AO1)	Accept processes for modify 1. and 2. Accept 'adds carbohydrate to' for modify 1. Accept 'adds lipid to' for modify 1. Accept lipoprotein for glycoprotein 1 or 2. Accept chylomicron for glycoprotein/glycolipid 2. Accept phospholipid for lipid Accept additional marking point, 4. Make/transport polysaccharides, but ignore cellulose

Question	Marking Guidance	Mark	Comments
03.3	Answer key: India, Europe, East Asia, Central America, South America	1 (AO3)	

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03.4	Correct answer of 8.3814 million/8 381 400/8.3814 x 10⁶ OR 8.4 million/8400000/8.4 x 10⁶;	1 (AO2)	Accept any correct numerical equivalent answer

Question	Marking Guidance	Mark	Comments
03.5	1. Mutation produced <u>allele</u>; 2. Those with allele/resistance less likely to/do not get malaria/<i>P vivax</i> OR Those with allele/resistance survive malaria/<i>P vivax</i>; 3. (So more likely to) reproduce and pass on the allele; 4. (Over generations) allele frequency increases;	4 (2 x AO1, 2 x AO2)	Reject gene for allele only once 1. Reject 'mutation caused by infection/exposure to <i>P. vivax</i>'. 2. Accept converse; eg 'people lacking the allele die from malaria' 4. Accept description of increasing frequency of allele eg 'higher proportion', 'more common' but 'ignore increase in number of allele'
